

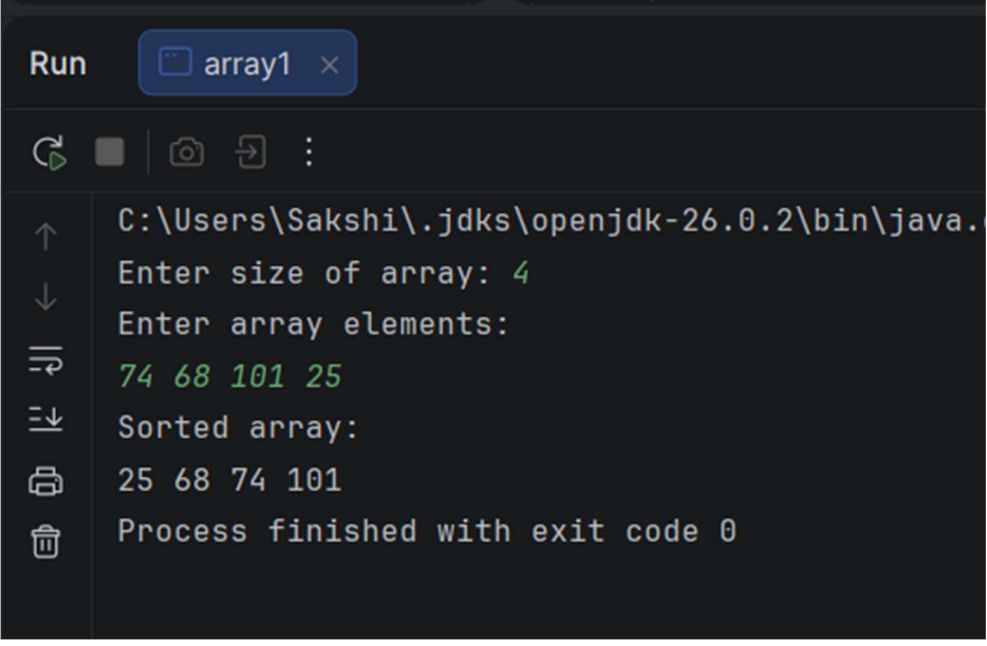
JAVA ARRAY ASSIGNMENT

Q2.

CODE:

```
import java.util.Scanner;
public class array1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for(int i = 0; i < n; i++) {
            for(int j = i + 1; j < n; j++) {
                if(arr[i] > arr[j]) {
                    int temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("Sorted array:");
        for(int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:



```
Run array1 x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.
Enter size of array: 4
Enter array elements:
74 68 101 25
Sorted array:
25 68 74 101
Process finished with exit code 0
```

Q3.

CODE:

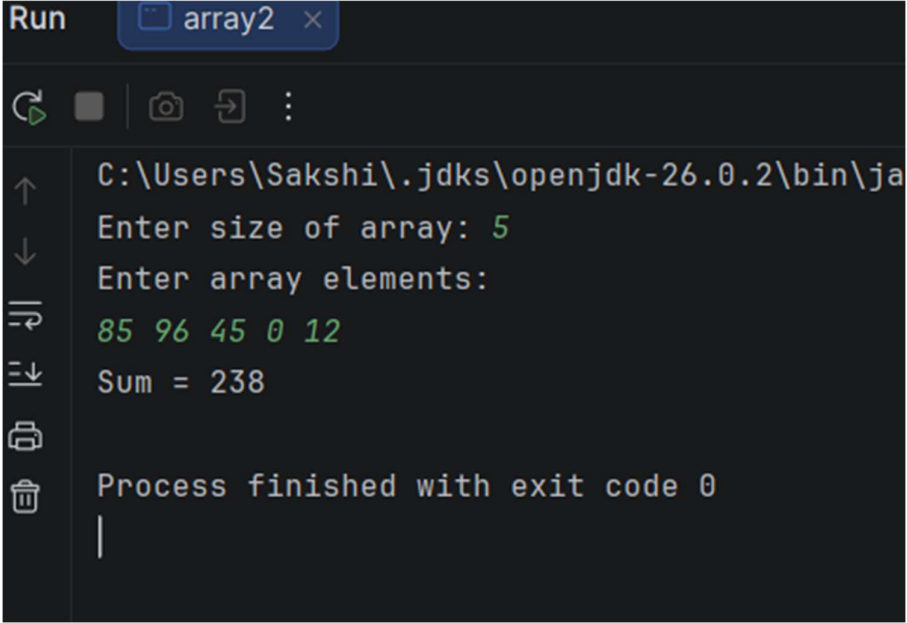
```
import java.util.Scanner;
public class array2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int sum = 0;
        for(int i = 0; i < n; i++) {
```

```

        sum = sum + arr[i];
    }
    System.out.println("Sum = " + sum);
    sc.close();
}
}

```

OUTPUT:



```

Run array2 x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\ja
Enter size of array: 5
Enter array elements:
85 96 45 0 12
Sum = 238
Process finished with exit code 0
|

```

Q4.

CODE:

```

import java.util.Scanner;
public class array3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
    }
}

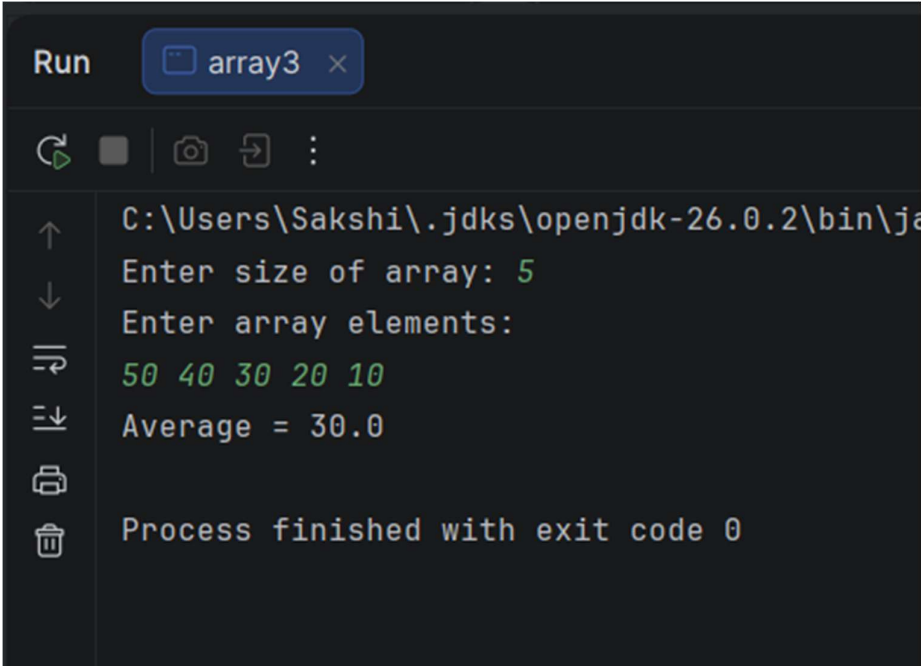
```

```

        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int sum = 0;
        for(int i = 0; i < n; i++) {
            sum = sum + arr[i];
        }
        double average = (double) sum / n;
        System.out.println("Average = " + average);
        sc.close();
    }
}

```

OUTPUT:



```

Run array3 x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java
Enter size of array: 5
Enter array elements:
50 40 30 20 10
Average = 30.0
Process finished with exit code 0

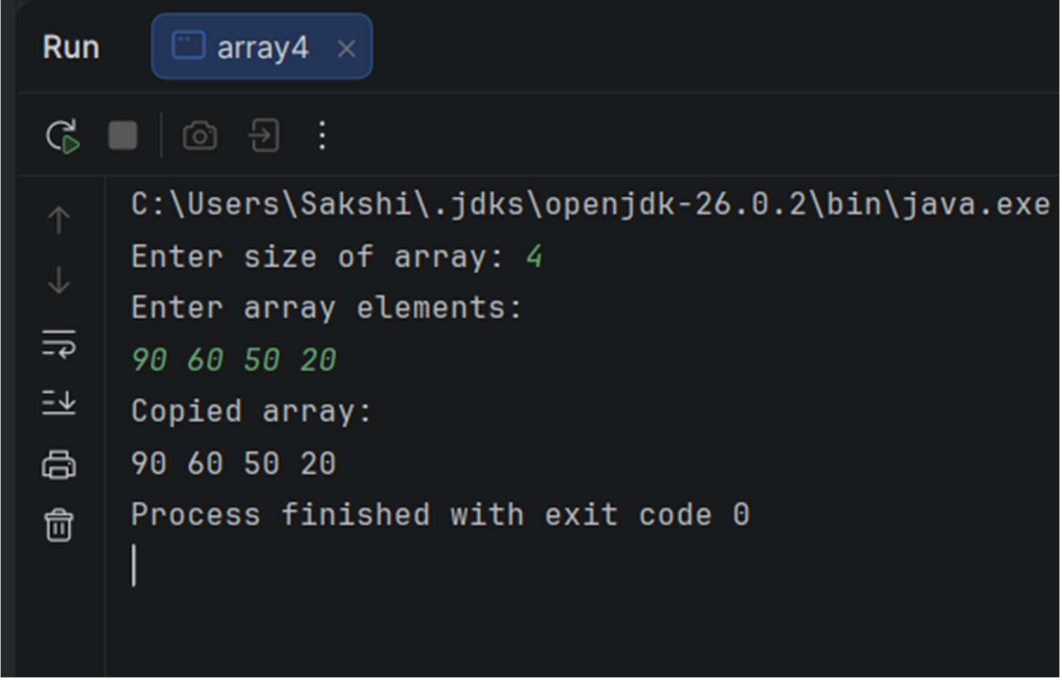
```

Q5.

CODE:

```
import java.util.Scanner;
public class array4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        int copy[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for(int i = 0; i < n; i++) {
            copy[i] = arr[i];
        }
        System.out.println("Copied array:");
        for(int i = 0; i < n; i++) {
            System.out.print(copy[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:



```
Run array4 x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe
Enter size of array: 4
Enter array elements:
90 60 50 20
Copied array:
90 60 50 20
Process finished with exit code 0
```

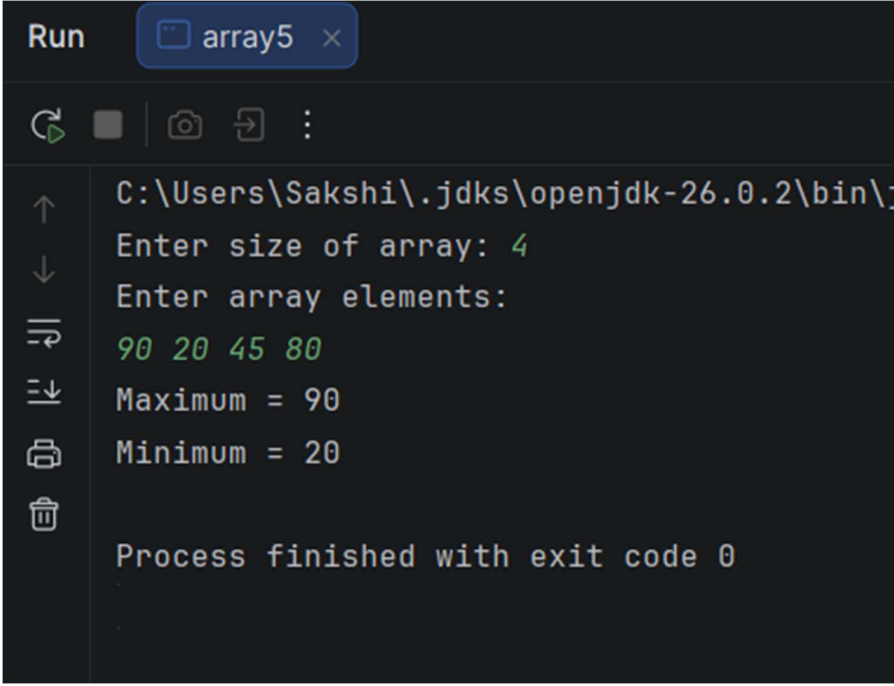
Q6.

CODE:

```
import java.util.Scanner;
public class array5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int max = arr[0];
        int min = arr[0];
        for(int i = 1; i < n; i++) {
            if(arr[i] > max) {
```

```
        max = arr[i];
    }
    if(arr[i] < min) {
        min = arr[i];
    }
}
System.out.println("Maximum = " + max);
System.out.println("Minimum = " + min);
sc.close();
}
}
```

OUTPUT:



The screenshot shows a 'Run' window titled 'array5' with a dark theme. The console output is as follows:

```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\j
Enter size of array: 4
Enter array elements:
90 20 45 80
Maximum = 90
Minimum = 20

Process finished with exit code 0
```

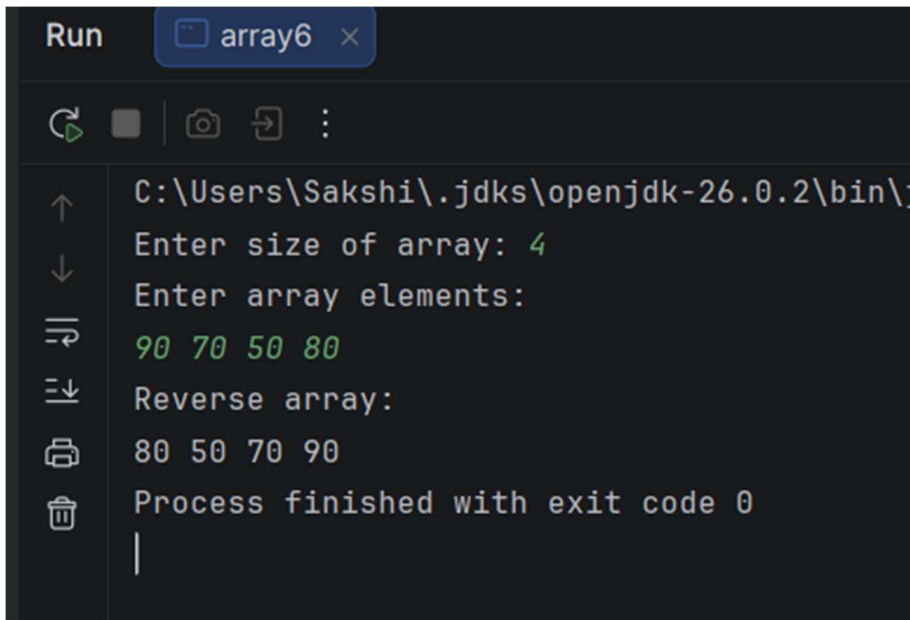
On the left side of the console, there is a vertical toolbar with icons for: up arrow, down arrow, undo, redo, copy, paste, and delete.

Q7.

CODE:

```
import java.util.Scanner;
public class array6 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("Reverse array:");
        for(int i = n - 1; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:



```
Run array6 x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe
Enter size of array: 4
Enter array elements:
90 70 50 80
Reverse array:
80 50 70 90
Process finished with exit code 0
```

Q8.

CODE:

```
import java.util.Scanner;
public class array7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("Duplicate values:");
        for(int i = 0; i < n; i++) {
            for(int j = i + 1; j < n; j++) {
                if(arr[i] == arr[j]) {
                    System.out.println(arr[i]);
                }
            }
        }
    }
}
```

```

    }
}
}
sc.close();
}
}

```

OUTPUT:

```

Run array7 x
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\
Enter size of array: 6
Enter array elements:
45 12 54 75 45 82
Duplicate values:
45
Process finished with exit code 0
|

```

Q9.

CODE:

```

import java.util.Scanner;
public class ComplexNumber {
    int number1;
    int number2;
    void setNumber1(int number1) {
        this.number1 = number1;
    }
}

```

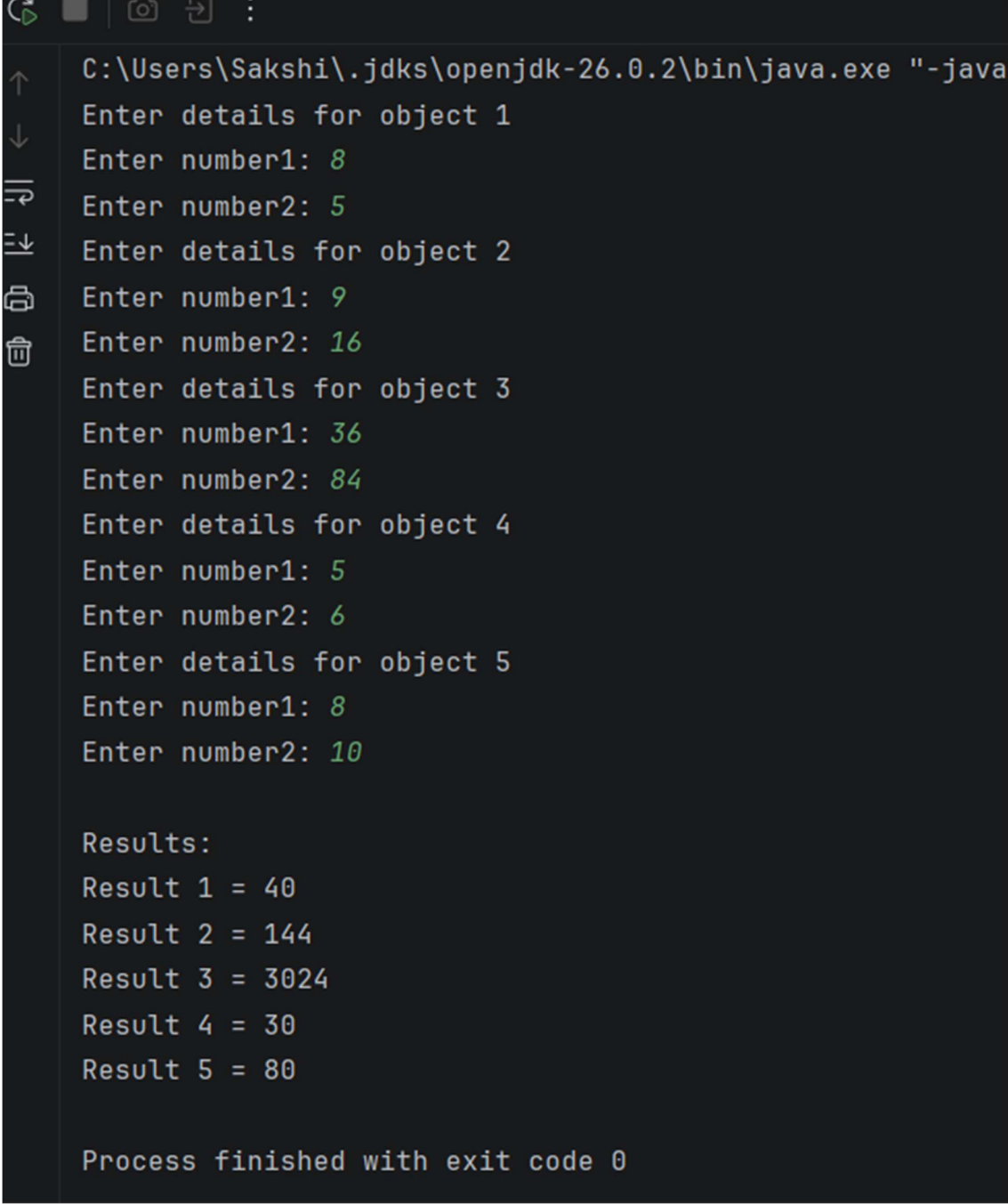
```

    }
    void setNumber2(int number2) {
        this.number2 = number2;
    }
    int getNumber1() {
        return number1;
    }
    int getNumber2() {
        return number2;
    }
    int computeComplexNumber() {
        return number1 * number2;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ComplexNumber arr[] = new ComplexNumber[5];
        for(int i = 0; i < 5; i++) {
            arr[i] = new ComplexNumber();
            System.out.println("Enter details for object " + (i + 1));
            System.out.print("Enter number1: ");
            int n1 = sc.nextInt();
            System.out.print("Enter number2: ");
            int n2 = sc.nextInt();
            arr[i].setNumber1(n1);
            arr[i].setNumber2(n2);
        }
        System.out.println();
        System.out.println("Results:");
        for(int i = 0; i < 5; i++) {
            int result = arr[i].computeComplexNumber();
            System.out.println("Result " + (i + 1) + " = " + result);
        }
    }
}

```

```
        sc.close();
    }
}
```

OUTPUT:

A screenshot of a Java IDE terminal window. The title bar shows the file path: C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe "-java. The terminal content shows the program's execution flow. It prompts for details for five objects, each requiring two numbers. The user enters values in green text. After the inputs, the program displays the results for each object, also in green text. The final line indicates the process finished with exit code 0.

```
C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin\java.exe "-java
Enter details for object 1
Enter number1: 8
Enter number2: 5
Enter details for object 2
Enter number1: 9
Enter number2: 16
Enter details for object 3
Enter number1: 36
Enter number2: 84
Enter details for object 4
Enter number1: 5
Enter number2: 6
Enter details for object 5
Enter number1: 8
Enter number2: 10

Results:
Result 1 = 40
Result 2 = 144
Result 3 = 3024
Result 4 = 30
Result 5 = 80

Process finished with exit code 0
```

Q10.

CODE:

```
import java.util.Scanner;
public class array8 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("Duplicate values:");
        for(int i = 0; i < n; i++) {
            for(int j = i + 1; j < n; j++) {
                if(arr[i] == arr[j]) {
                    System.out.println(arr[i]);
                }
            }
        }
        sc.close();
    }
}
```

OUTPUT:

Run array8 x








C:\Users\Sakshi\.jdk\openjdk-26.0.2\bin
Enter size of array: 5
Enter array elements:
25 48 95 75
48
Duplicate values:
48

Process finished with exit code 0